

VR Competency-Based Pilot Training for Threat and Error Management

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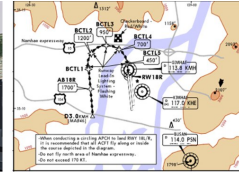
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Training With AI Simulated Pilot



Virtual Reality Cockpit



Busan Circling Approach

Abstract

Full Flight Simulators are full-motion, cockpit-replica systems that serve as the industry standard for pilot training. However, their high operating cost and limited availability restrict practice opportunities for rare yet critical procedures. In contrast, VR technology delivers a scalable alternative at orders of magnitude lower cost. In partnership with Singapore Airlines (SIA), we developed a VR prototype that recreates the Busan visual approach scenarios with

injected threats and errors to enable self-directed learning, allowing pilots to train on these complex scenarios anytime and anywhere.

CCS Concepts

• **Human-centered computing** → **Virtual reality**.

Keywords

Virtual Reality, Aviation, Pilot Training, Multi-modal Interaction

ACM Reference Format:

Teong Leong Chuah, Brian Soon Wei Chiam, Lindy Li Wen Lim, Ahmad Iqbal Othman, Jia Wang Tay, Vinh-Thuyen Nguyen-Truong, Catherine Wan Ting Leo, Leslie Siew Mun Chong, Jussi Keppo, and Eng Tat Khoo. 2025. VR Competency-Based Pilot Training for Threat and Error Management. In *SIGGRAPH Asia 2025 XR (SA XR '25)*, December 15–18, 2025, Hong Kong, Hong Kong. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3761667.3761949>

1 Introduction and Background

Prior research on VR flight simulators (VRFS) has primarily focused on design elements, cockpit interactions and familiarization [Auer

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SA XR '25, Hong Kong, Hong Kong

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ACM ISBN 979-8-4007-2168-7/25/12

<https://doi.org/10.1145/3761667.3761949>

et al. 2023, 2021; Oberhauser et al. 2018], leaving a gap for Threat and Error Management (TEM) - an essential part of a commercial pilot's biannual Evidence-Based Training (EBT). This gap exists even in studies with scenarios or traffic patterns [Oberhauser et al. 2016]. In the industry-standard Competency-Based Training and Assessment (CBTA) framework [IATA 2024], *threats* are existing conditions outside of the pilot's control while *errors* are pilot-induced deviations [Maurino 2005]. VR has demonstrated potential in enhancing training across fields [Liu et al. 2024; Ng et al. 2022], indicating opportunity for aviation applications. In this research, we focus on designing VR flight simulators (VRFS) as cost-effective and portable supplementary solutions for TEM training. Our study focuses on the visual approach into Gimhae International Airport (PUS), selected for its operational relevance to our industry partner and the complexity of the procedure, further evidenced by another airline's recent PUS incident [Aviation Safety Network 2025]. The Busan visual approach has a high workload where pilots are responsible for maintaining terrain and obstruction avoidance until landing.

2 Design and Implementation

The system comprises a Boeing 737-MAX 8 cockpit model, terrain surrounding airport, hand and eye tracking support with the Meta Quest Pro's in-built capabilities, novel VR cockpit interactions such as a pull-to-rotate flaps lever, and an AI Simulated Pilot which acts as the captain and Pilot Flying (PF) in the circling approach scenarios. The PF is instructional and corrective, based on a behavior tree system built around two scenarios, and is a proof-of-concept that can be extended and improved upon to include other scenarios, threats, and errors. The PF tests the user by failing to handle a threat or intentionally committing an error, and the user, acting as Pilot Monitoring (PM), can recognize and resolve these through voice and hand interaction. At the end of each scenario, the user receives text feedback on their performance, highlighting the system's potential for self-directed review of safety-critical scenarios.

The scenarios were tailored in collaboration with Singapore Airlines instructor pilots (20+ years of commercial flying experience), who regularly tested and provided feedback on replicated VR cockpit instruments, validated flight path and behaviors, while ensuring scenarios avoided negative training. We aimed to develop scenarios for *competency-based* training, where the recognition and resolution of threats and errors are linked to competencies and observable behaviors (OBs) from the CBTA framework [IATA 2024].

Our system's CBTA-based *threats and errors* are primarily lateral and vertical deviations, proceeding towards the wrong runway, failure to go-around after destabilization on approach, and unstable approach. Others that could emerge in the scenario include, but are not limited to, normal checklist errors, callout errors, and communication errors between pilots [IATA 2024]. Only the primary threats and errors are recognized by the system in the prototype's current iteration, while others would still require an observing instructor.

3 User Experience / Application

The prototype's overall user experience allows users to get a novel glimpse into commercial pilot training protocols. Users start with a *tutorial* that familiarizes them with the VR environment and cockpit interactions. Our prototype features two different scenarios of the visual (circling) approach into PUS. In *the first scenario*, the PF intentionally forgets to engage lateral navigation (LNAV) at the

correct time, resulting in the aircraft heading for dangerous terrain, and later lines up with the wrong adjacent runway. The user is expected to call out the LNAV error, identify the wrong runway, and call for a go-around. *The second scenario* features the same LNAV error, but the other error is the PF setting an insufficient rate of descent, leading to an aircraft on high profile, ie. too high for a safe landing. The PM is expected to call out high profile, an increase in the rate of descent, and a go-around after checking for stabilization criteria at 500 feet. The user's proficiency as PM is tested by monitoring cockpit and external conditions, using proper protocol to recover from an Undesired Aircraft State (UAS), and escalating communication if required.

4 Discussion and Future Work

The current system's feedback is limited to threat and error handling. The next step is to extend capabilities to more OBs including those related to communication aspects (e.g. clarity and assertiveness). We are developing pilot behavior recognition on Full Flight Simulator (FFS) data to support VR feedback system. We are in the midst of evaluating the prototype through a three-arm randomized controlled trial (FFS, VR, no-training) on a group of type rated 737 pilots; results will be reported in a separate publication.

Acknowledgments

This research is supported by National Research Foundation, Singapore and A*STAR, under its RIE2020 Industry Alignment Fund – Industry Collaboration Projects (IAF-ICP) grant call (Grant No. I2001E0059) – SIA-NUS Digital Aviation Corp Lab. We would like to thank SIA Captains Adrian Amaladoss, Daniel Gan, Davinson Sim, David Hawkes, Gabriel Goh, Sean D'Penha, Daniel Dickson, Christopher Chua and Senior First Officer Daryl Wong.

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